



UNIVERSITÀ DI PARMA

LABORATORY #6

- Perspective transformations
- We want to find image coordinates of world points

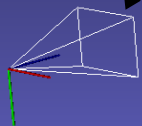
- Given
 - A point cloud $\rightarrow$ collections of world points with world coordinates
 - Scan.dat
 - Camera(s) parameters $\rightarrow$ intrinsic and extrinsic parameters
 - params_*.dat
- Generate resulting image(s)

- Can be used for data visualization
- Difficult to install
- If not available
 - Comment out the `#define USE_OPENCVVIZ` in `utils.h`
 - Use `gnuplot` to visualize data:
 - `gnuplot`
 - `splot "scan_gnuplot.dat"`

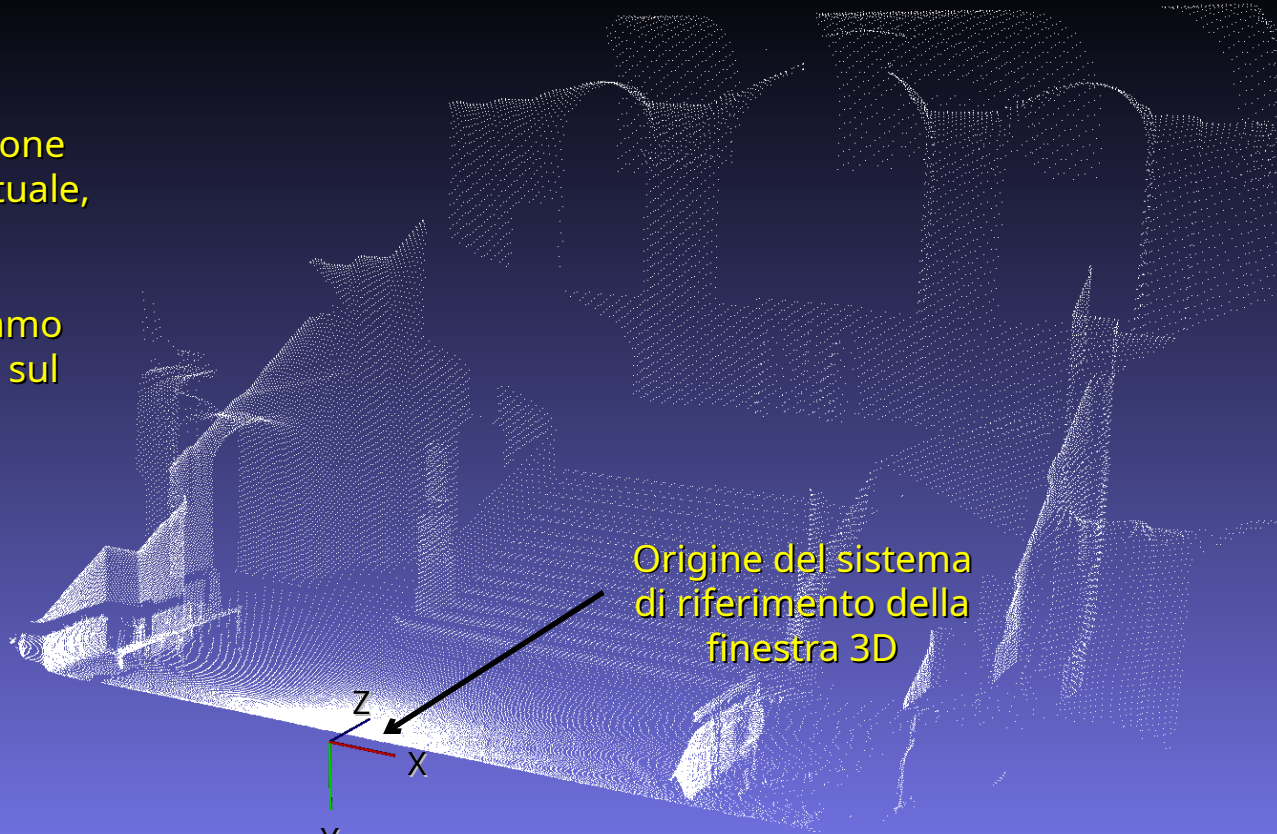
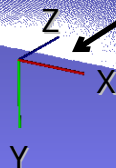
- Each line of scan.dat corresponds to a 3D world point as
 - -0.0434742 -4.82982 0.499645
 - 0.0295245 -4.82834 0.541775
 - 0.103245 -4.84538 0.584323
 - 0.176457 -4.84095 0.626577
 - ...

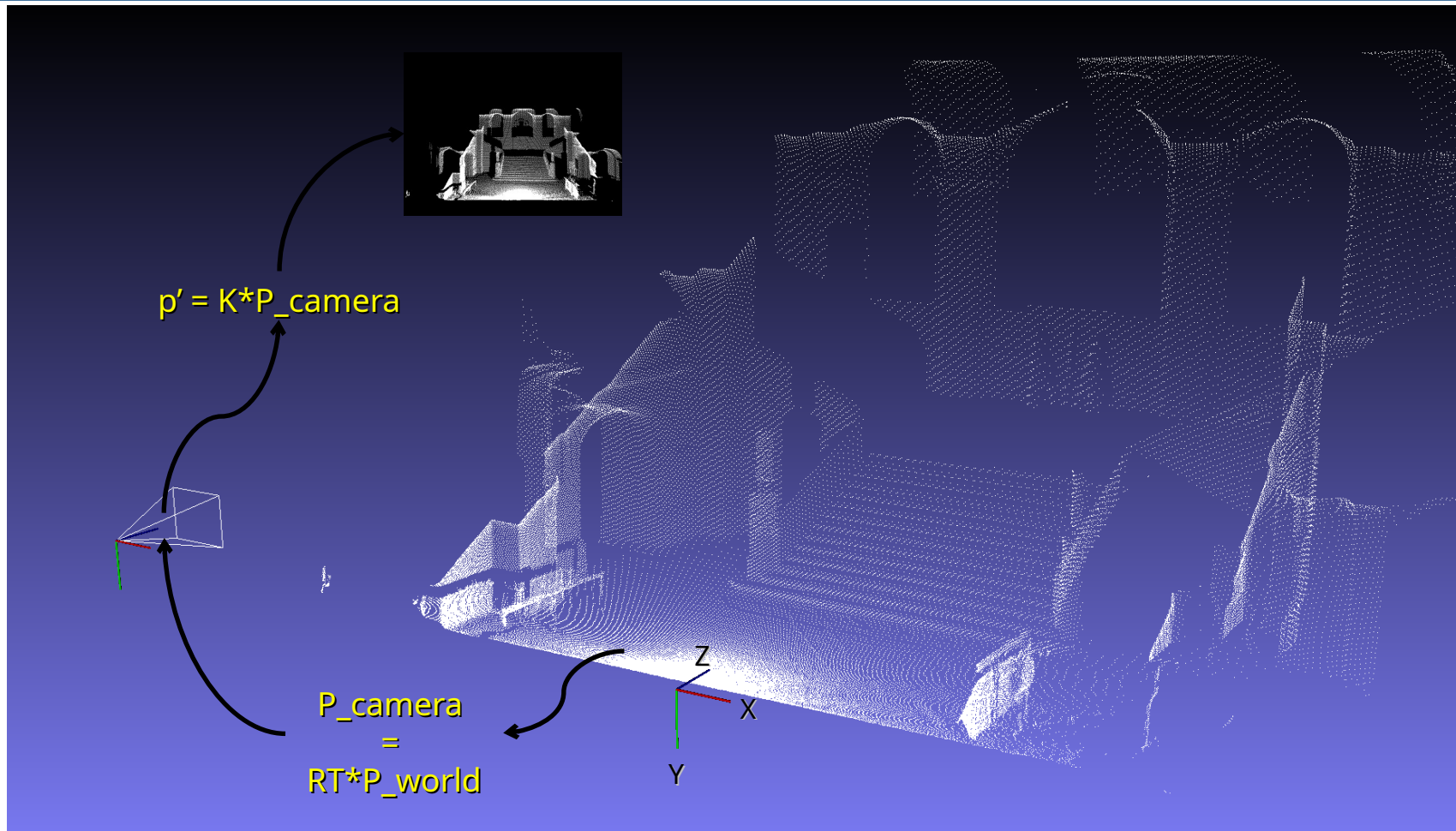
- Params_*.dat files contains camera data as follows:
 - 640 480 //width and height
 - 400 400 //f (pixels)
 - 320 240 //optical center u0, vo
 - 0.0 0.0 0.0 //orientation wrt x,y,z
 - 0.0 -5.0 -10.0 //position wrt x,y,z

Posizione ed orientazione
della nostra camera virtuale,
definita dentro
params_front.dat
E' quella su cui dobbiamo
proiettare in punti 3D sul
piano immagine



Origine del sistema
di riferimento della
finestra 3D





- Rotate camera around the center of gravity of the cloud point
 - Also rotate camera around Y axis to always frame the data
- Generate resulting images